

ANANYA SAHA

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• 2026-Present

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RESEARCH INTERESTS

Image and Video Analysis, Attention Mechanisms and Model Explainability, Few-Shot Learning, Multi-Modal Fusion and Meta-Learning, Clinical Diagnostics AI. Broad interest in translating advanced machine learning techniques into practical diagnostic tools for resource-limited settings.

EDUCATIONAL BACKGROUND	ACADEMIC QUALIFICATIONS
MTech in CSE (Artificial Intelligence and Data Science Specialization) •Jadavpur University •2024-2026 •Grade: 9.51 CGPA	NET ,Dec 2025, CSE Qualified: PHD, Assistant Professor
BTech in CSE •Techno Main SaltLake •2020-2024 •Grade: 9.38 CGPA	GATE ,2024, CSE Qualified: 591 score
Higher Secondary School (XII) •Indira Gandhi Memorial High School •2019-2020 •Grade: 92%	YGPA ,2023-2024, (4th Year) Position: 1, YGPA:9.45
Secondary School (X) •Julien Day School, Kalyani •2017-2018 •Grade: 93.75%	National Cyber Olympiad ,2017 City Rank: 2, Zonal Rank(WB): 78 International Rank : 549

RESEARCH EXPERIENCE

ACCEPTED/PUBLISHED

April 2026	Few-Shot Learning based Multimodal Skin Lesion Classification Authors: Saptarshi Pani*, Ananya Saha*, Ram Sarkar [*equal contributions] <u>ACCCEPTED</u> in <i>PHAROS-AIF-MIH Workshop</i> , held in conjunction with CVPR 2026 Developed a dual-stream few-shot learning framework integrating visual and textual modalities for skin lesion classification under limited labeled data. The vision stream combines frozen CLIP ViT-B/16 and DenseNet-121 encoders with a Gaussian fuzzy adaptive fusion using learnable confidence weights based on activation magnitudes. Designed a novel alignment mechanism between prompt-tuned CLIP text encoder (learnable context tokens) and a trainable metadata encoder, projecting both into shared CLIP embedding space for logit-level fusion. Evaluated on PAD-UFES-20, HAM10000, and ISIC datasets in 1–10 shot settings, achieving competitive accuracy with reduced supervision.
March 2026	CoMFDNet: A Correlation-guided Meta-Fusion based Dual Network for Glaucoma Classification Authors: Ananya Saha, Gouranga Maity, Ram Sarkar <u>PUBLISHED</u> in <i>NHIB (Springer)</i> Proposed CoMFDNet, a dual-network framework for automatic glaucoma classification from retinal fundus images. The model combines ResNet50 and MobileNetV2 feature extractors with a cosine similarity–based correlation fusion module, enhanced by channel and spatial attention mechanisms. A meta-learner adaptively balances both branches to improve robustness. Evaluated on ACRIMA and LAG, the approach achieves up to 99.63% accuracy, demonstrating strong performance for automated clinical glaucoma screening.
June 2025	NeuralCodOpt: Codon optimization for the development of DNA vaccines Authors: Tapan Chowdhury, Aishwarya Saha, Ananya Saha, Arnab Chakraborty, Nibir Das <u>PUBLISHED</u> in <i>Computational Biology and Chemistry (Elsevier)</i> Developed NeuralCodOpt, a machine learning–based algorithm for organism-specific codon optimization to improve gene translation efficiency. The framework combines K-Means clustering to estimate target Codon Adaptation Index ranges and reinforcement learning to automate sequence optimization. Evaluated across 130 species, the method achieves 86.7% accuracy and demonstrates potential to enhance protein expression in DNA expression vectors, supporting applications in gene therapy and DNA vaccine development.

May 2023

Commercial aspect of Telemedicine in Indian scenario

Authors: Prof. Jafor Ali Akhan, Ananya Saha, Shubhankar Ghosh, Dr. Himadri Nath Saha
PUBLISHED in IJESI

Analyzed the evolution and impact of telemedicine in healthcare delivery, particularly in low- and middle-income countries such as India. The study highlights the expansion of SATCOM-based telemedicine networks initiated by Indian Space Research Organisation and the rapid adoption of telehealth services during the COVID-19 pandemic by the Ministry of Health and Family Welfare. Findings emphasize telemedicine's potential to improve healthcare accessibility and reduce system burden in resource-constrained settings.

March 2023

Heart Disease Treatment in Telemedicine using Internet of Things

Authors: Prof. Jafor Ali Akhan, Ananya Saha, Shubhankar Ghosh, Dr. Himadri Nath Saha
PUBLISHED in IJESI

Presented at the 5th Regional Science and Technology Congress, Region-4 (Kolkata, Howrah, South 24 Parganas), 2022–23.

[Abstract published in the Conference Book of Abstracts]

Proposed a low-cost Internet of Things-based telemedicine system for real-time heart disease monitoring. The framework uses a fingertip-based heart rate sensor to detect and transmit patient heartbeat signals wirelessly to a computer or Android application for visualization and analysis. The system enables continuous remote monitoring, alerts, and feedback to help detect potential cardiac risks and improve telemedicine-based healthcare management.

ONGOING

March 2026

GILF-Net: Gated Iterative Learnable Lifting Fusion for Dual-Branch Attention-Based Medical Image Classification

Supervisor: Prof. Ram Sarkar, JU

Developed a dual-encoder deep learning architecture with a novel gated iterative lifting fusion mechanism for medical image classification. The framework combines MobileNetV2-SE and DenseNet201-CBAM encoders, projects features into a shared 512-dimensional latent space via 1×1 convolutions, and performs bidirectional iterative refinement using learnable residual updates and spatial gating. The fusion module dynamically exchanges complementary information between branches across T iterations, progressively enhancing discriminative representations. Evaluated on LAG (glaucoma), HAM10000 (skin lesions), and PBC (blood cells) datasets, achieving superior accuracy over single-branch baselines and conventional fusion methods.

Feb 2026

GI-GEM: Geometry-Influenced Explainability for Deep Vision Models

Supervisor: Prof. Ram Sarkar, JU

Proposed GI-GEM (Geometry-Influenced Gradient Explanation Maps), a model-agnostic explainability framework that analyzes geometric sensitivity of deep vision models using controlled affine and elastic transformations. The method fuses geometric influence maps with gradient-based activations to generate structure-aware explanations without retraining. Evaluated on ImageNet, 3DIRCADb, and JUVDSi v1, the approach produces sharper localization and improved faithfulness compared to Grad-CAM and LIME across multiple vision tasks.

Feb 2026

Contrastive Training and Multi-Prompt Distillation based Few-Shot Learning for Skin Lesion Classification

Supervisor: Prof. Ram Sarkar, JU

Proposed a three-stage few-shot learning framework for skin lesion classification under limited labeled data. The pipeline integrates contrastive image–text pretraining, prompt-consistency distillation with cross alignment and knowledge distillation with frozen teacher, and prototypical few-shot learning using a CLIP ViT-B/32 backbone to learn domain-aware visual–semantic representations. Evaluated on HAM10000, ISIC 2018, and BreakHis, the method achieves strong performance across multiple few-shot settings (1–10 shots), demonstrating improved robustness for medical image classification in data-scarce scenarios.

March 2026

Few-Shot Breast Ultrasound Lesion Segmentation using Dual-Encoder and GA-based Channel Pruning

Supervisor: Prof. Ram Sarkar, JU

Proposed a novel CAGSC fusion module and evolutionary channel pruning strategy for prototypical few-shot medical image segmentation, designing a dual-encoder architecture that integrates a fine-tuned FastSAM encoder with a trainable Swin-Attention U-Net, enriched with dense cross-scale skip connections for multi-level structural context at the decoder; employed gradient-free GA-based structured channel selection across four feature pathways to jointly optimize segmentation accuracy and model compactness, followed by post-pruning fine-tuning for stability, and validated the framework on three public ultrasound datasets with state-of-the-art Dice scores (83.91%–91.83%) under 1/3/5-shot settings.

SKILLS

- **Programming & Development:** Python, C, Web Development (HTML, CSS, JavaScript)
- **Emerging Technologies:** Machine Learning, Deep Learning, Internet of Things (IoT)
- **Research & Academic Skills:** Literature Review, Scientific Writing, Content Writing
- **Professional Skills:** Teamwork, Leadership, Laboratory Assistance, Student Mentoring
- **Languages:** Bengali (Native/Fluent), English (Professional Proficiency), Hindi (Limited Proficiency)

TEACHING/MENTORING EXPERIENCE

Present

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March
2026

Lab Assistant, of lab UG/CSE/ES/S/123 (Computer Programming Lab)

📍 **Jadavpur University**

Assisted Prof. Ram Sarkar in conducting Computer Programming laboratory sessions for two groups of 1st-year B.Tech CSE students (~60 students total, 30 per group). Delivered supplementary lectures on core C programming concepts, administered weekly programming assignments, and provided live debugging and problem-solving support during lab sessions. Evaluated student code implementations and maintained attendance records.

Present

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Oct 2025

Student Mentor, CMATER Lab

📍 **Jadavpur University**

Mentored undergraduate research intern under Prof. Ram Sarkar's supervision. Guided training in transfer learning implementations, multi-modal fusion techniques, and deep learning frameworks (PyTorch/TensorFlow). Provided instruction on research methodology including literature review, experimental design, result analysis, and academic writing. Supervised hands-on projects involving medical image classification and model evaluation, fostering development of technical and scholarly skills for graduate-level research.

Dec 2025

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Aug 2025

Lab Assistant, of lab CSE/PC/B/S/311 (Computer Graphics Lab)

📍 **Jadavpur University**

Assisted Prof. Ram Sarkar in conducting Computer Graphics laboratory sessions for three groups of 3rd-year B.Tech CSE students (~90 students total, 30 per group). Delivered supplementary lectures on core graphics concepts, designed and administered weekly programming assignments, and provided live debugging and problem-solving support during lab sessions. Evaluated student code implementations, maintained attendance records, and invigilated internal laboratory examinations. Supervised and mentored student mini-projects on graphics algorithms and visualization techniques.

REFERENCES

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